REFRAME: Reference-Genome Reassessment for Accurate Mutation Evaluation

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Abstract

Mapping sequencing reads to a reference genome and calling variants are essential preprocessing steps in genomic and epigenomic analysis. However, multiple human references exist, and no standardized framework determines which is most appropriate. Because variant calls represent sample-reference differences, variant detection is inherently reference-dependent. Differences in completeness, ancestry representation, and structure can introduce bias and influence mapping accuracy, variant detection, and interpretation.

This project evaluates how reference choice affects whole-genome sequencing alignment and variant detection across three linear references: hg19, hg38, and T2T-CHM13. hg38 is an updated version of hg19, whereas T2T-CHM13 is a telomere-to-telomere reference that resolves previously missing or ambiguous regions. The Human Pangenome, a graph-based reference constructed from diverse haplotypes, was excluded because it requires a different alignment framework and is less interchangeable with linear references.

Three related whole-genome sequencing samples were analyzed as a family dataset, providing a shared genetic baseline for evaluating reference-dependent differences while reducing unrelated inter-individual variation. In parallel, 39 sarcoma samples were processed to determine whether these differences persisted in a cancer-relevant cohort.

Paired-end reads were aligned to hg19, hg38, and T2T-CHM13 using BWA-MEM. BAM files were processed with samtools, variants called with DeepVariant, and calls filtered with bcftools to retain high-quality calls (depth >10x). To standardize coordinates, hg19 and hg38 VCF files were lifted over to CHM13v2 using Picard LiftoverVcf, enabling identification of shared and reference-specific variants.

Across datasets, mapping rates exceeded 99.7%, but resulting variant calls differed among references. hg19 and hg38 showed the strongest concordance, whereas T2T-CHM13 produced distinct profiles, including allele frequency shifts, heterozygous-to-homozygous ratios, and chromosome-specific distributions. More than 70% of variants were shared, but reference-specific differences persisted, especially in complex regions.

These findings demonstrate that reference choice influences variant detection and interpretation and should be carefully considered when developing, validating, and benchmarking genomic analysis pipelines.

Introduction/Background

Objective: Determine whether reference genome selection affects cfDNA variant detection and whether newer references recover variants missed by older references

Ref	Length	Diff (vs. hg19)	Year	# of Part.	Method of Assembly	Participant Demographic
hg19	3,095,693,983 bp (~3.10 Gb)	0	2009	~12	Clone-based Sanger sequencing with BAC-by-BAC assembly	Mostly one anonymous male donor, (mixed African-European ancestry)
hg38	3,088,286,401 bp (~3.09 Gb)	-7,407,582 bp	2013	~10	Updated assembly with corrected regions & alternate loci (NGS)	Primarily based on the same RP11 male donor background, with additional participants
T2T-CHM13	3,110,243,433 bp (~3.11 Gb)	+14,549,450 bp	2022	1	Long-read assembly using PacBio HiFi and Oxford Nanopore data	Derived from CHM13, a female hydatidiform mole cell line; no Y chromosome

Fig 1. Reference genome comparison. Summary of hg19, hg38, and T2T-CHM13 reference genome construction, length, participant source, and assembly approach

Methods & Materials

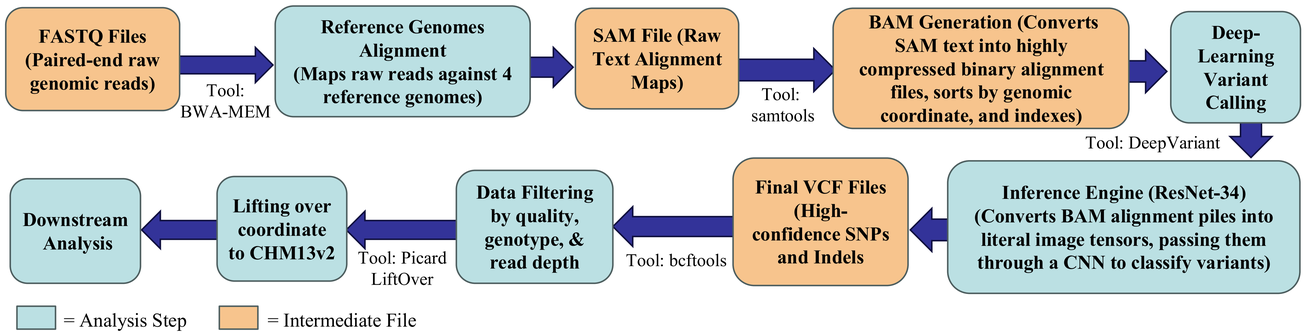


Fig. 2a: Bioinformatics pipeline executed via Stanford's Sherlock Cluster

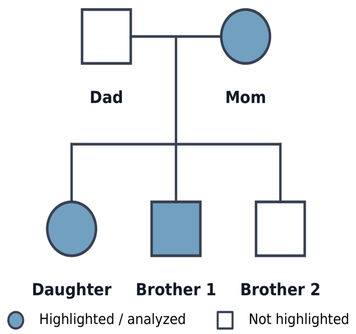


Fig. 2b: Pedigree representing initially studied family dataset

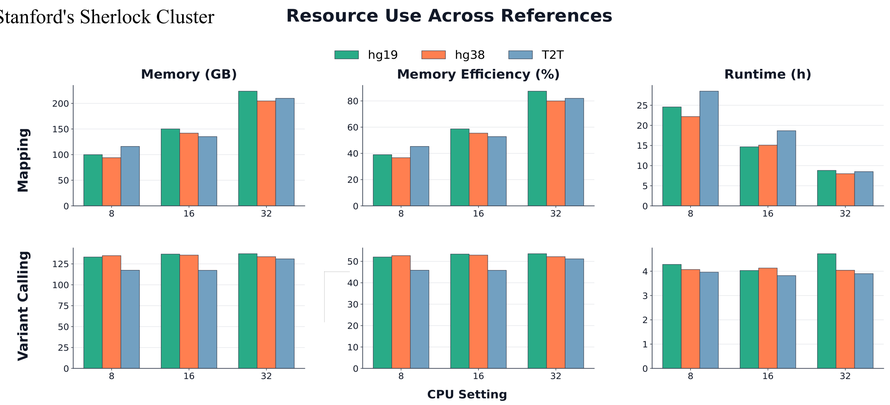
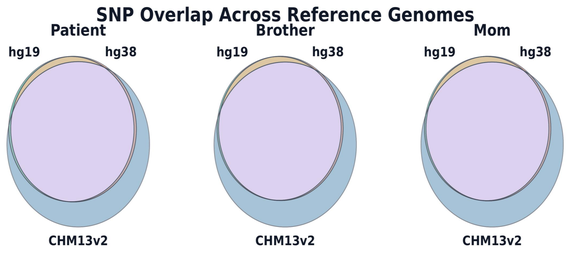


Fig. 2c: Resource Use Summary for alignment & variant calling across reference genomes (hg19, hg38, & T2T) on various CPU Settings (8, 16, & 32)

Results



Overlap Category	Patient	Brother	Mom
Shared by all 3	71.0%	70.2%	70.7%
hg19 & hg38 only	1.8%	2.1%	2.0%
hg19 & T2T only	0.2%	0.2%	0.2%
hg38 & T2T only	1.0%	0.9%	0.9%
hg19 only	1.2%	0.8%	0.8%
hg38 only	0.8%	0.6%	0.7%
T2T only	23.9%	25.1%	24.7%

Fig. 3b: Proportional Venn diagrams and summary percentages show shared and reference-specific SNPs across hg19, hg38, and CHM13v2 after coordinate standardization.

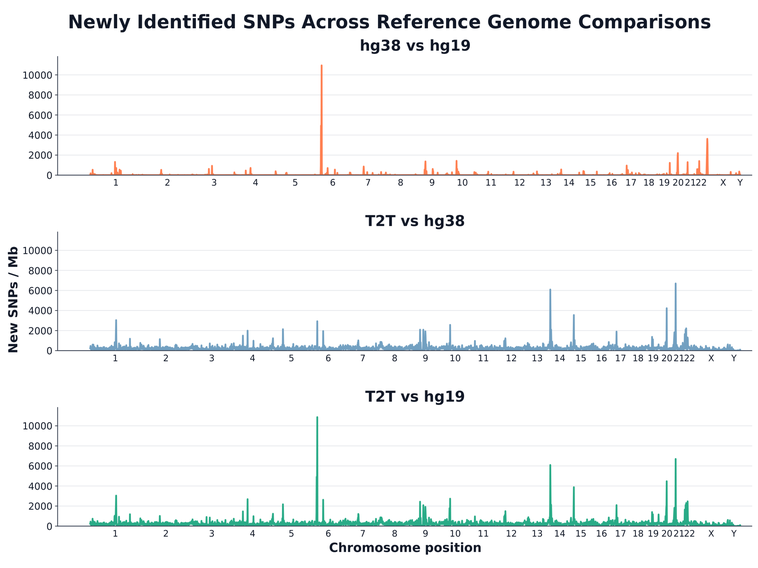


Fig. 4: Genome-wide SNP gain profiles show the number of SNPs detected in the newer reference but absent from the older comparison reference across chromosomal positions.

Results (Continued)

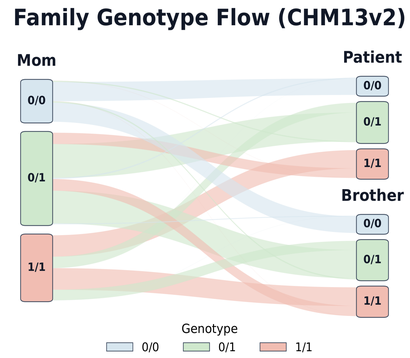
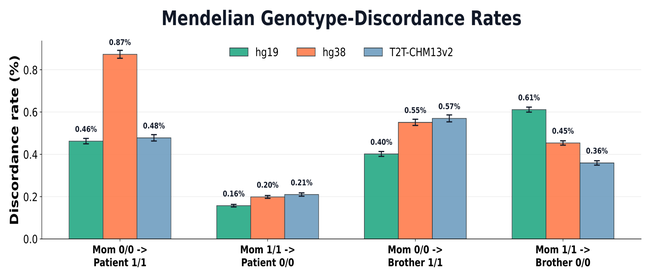


Fig. 5: Sankey plot shows maternal genotype transitions to the patient and brother in CHM13v2 coordinates; Bar plots quantify Mendelian genotype-discordance rates across references; error bars show Wilson 95% confidence intervals



Conclusions & Future Directions

- Successful mapping and matched variant-calling conditions still produced significantly different variant calls across reference genomes.
- hg19 and hg38 were most concordant; T2T-CHM13 showed distinct allele-frequency and chromosome-specific patterns.
- Reference-specific SNPs persisted mainly in complex genomic regions despite broad overlap across references.
- Next steps include completing sarcoma cfDNA and Human Pangenome analyses to test these patterns in cancer-relevant data.

Selected Sources

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